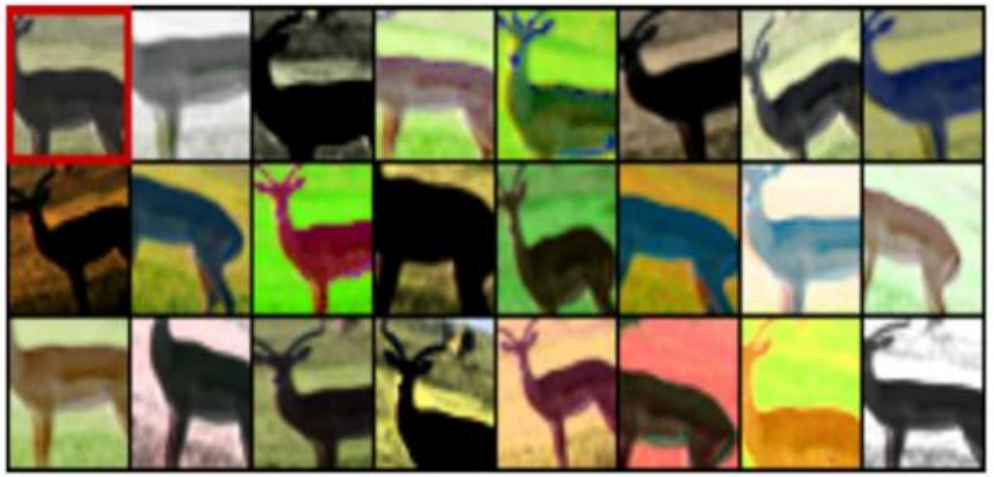
Some Examples Self-supervised Learning



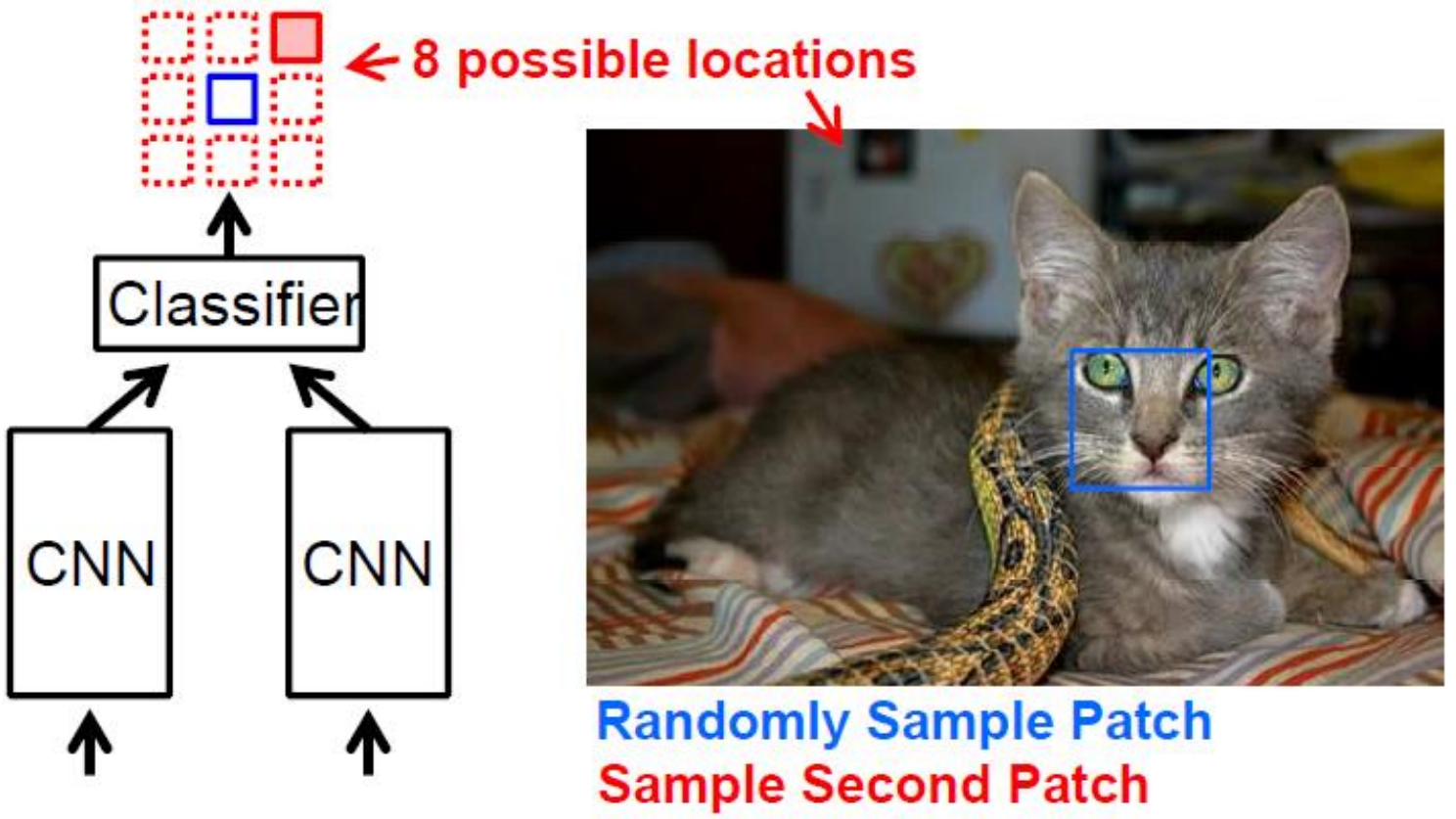
Example 1: Exemplar Networks

- Perturb/distort image patches through cropping and affine transformations
- Train a network to classify these exemplars as the same class



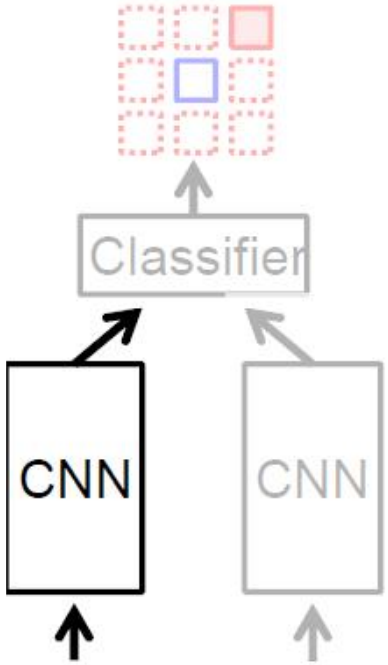
Dosovitskiy et al. “Discriminative Unsupervised Feature Learning with Exemplar Convolutional Neural Networks”, PAMI 2014

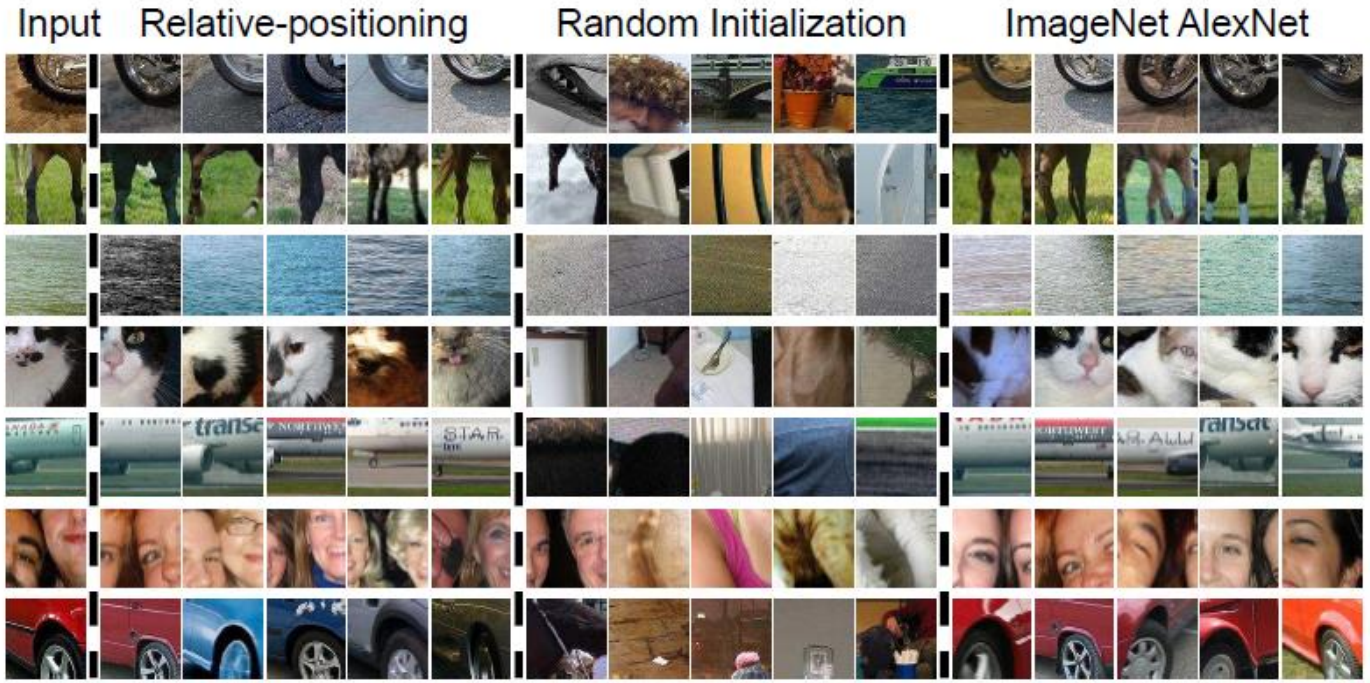
Example 2: Find relative position of patches



[Unsupervised visual representation learning by context prediction, Carl Doersch, Abhinav Gupta, Alexei A. Efros, ICCV 2015]

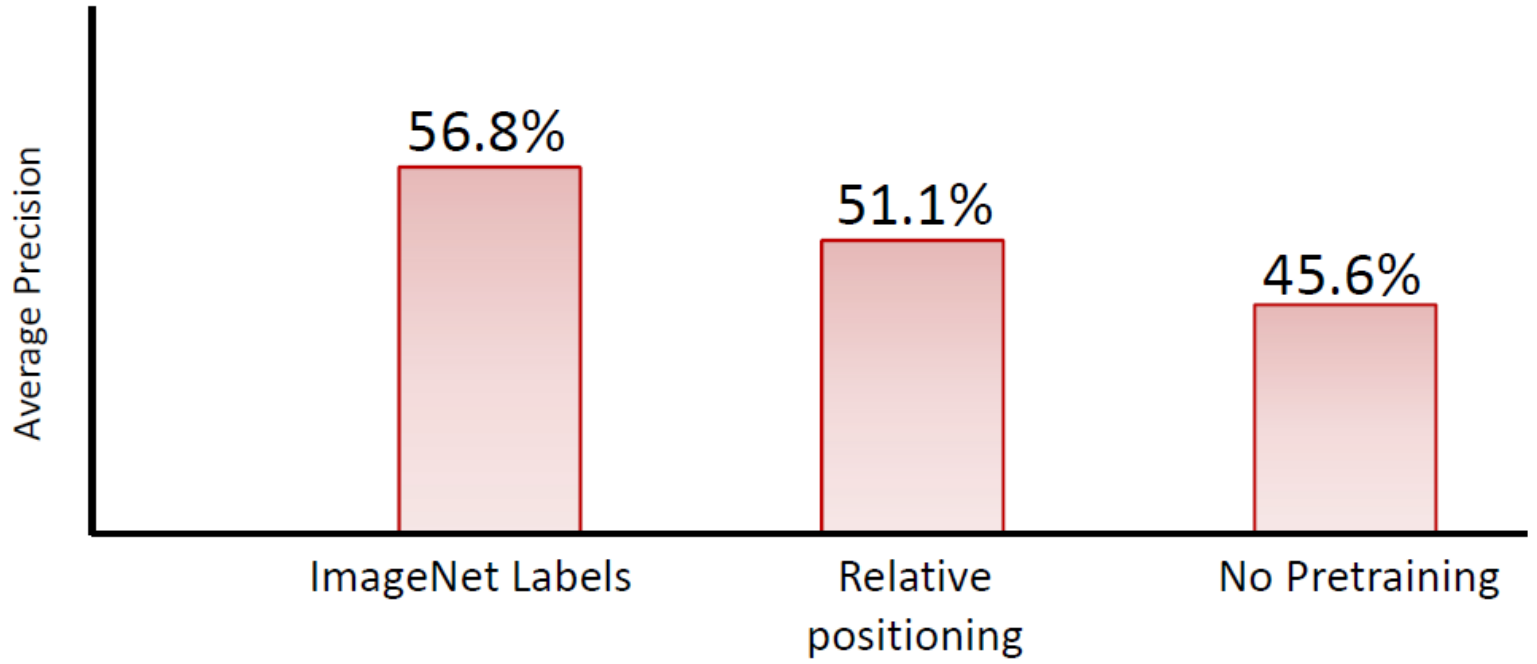
Example 2: Find relative position of patches





[Unsupervised visual representation learning by context prediction, Carl Doersch, Abhinav Gupta, Alexei A. Efros, ICCV 2015]

Example 2: Find relative position of patches



[Unsupervised visual representation learning by context prediction, Carl Doersch, Abhinav Gupta, Alexei A. Efros, ICCV 2015]

Reflection/Discussion point

Why does (should) self-supervised learning work?

e.g., for images, why should (does) it learn to extract features that are helpful for predicting objects, scenes, actions, etc.?



3 minutes

Self-Supervised Learning

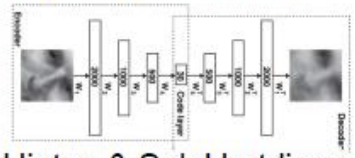
The network, through the proxy self-supervised task, is forced to learn what we really care about, e.g., semantic classes (what is a cat? what parts is it consist of?), in order to solve the task.



Self-Supervised Learning

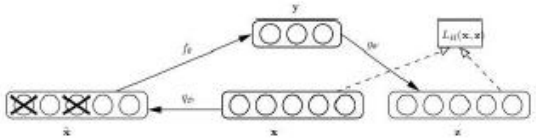
and many other creative approaches!

Autoencoders



Hinton & Salakhutdinov. Science 2006.

Denoising Autoencoders



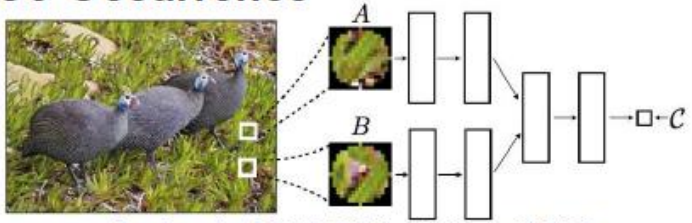
Vincent *et al.* ICML 2008.

Exemplar networks



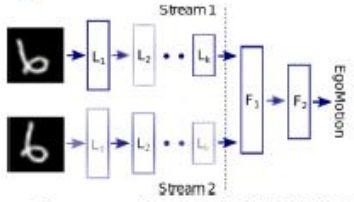
Dosovitskiy *et al.*, NIPS 2014

Co-Occurrence



Isola *et al.* ICLR Workshop 2016.

Egomotion

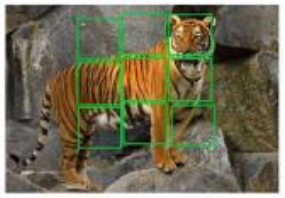






Agrawal *et al.* ICCV 2015 Jayaraman *et al.* ICCV 2015

Context



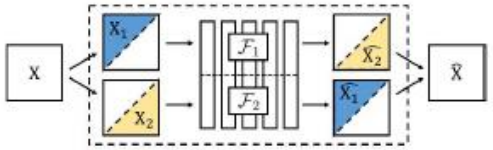


Noroozi et al 2016



Pathak *et al.* CVPR 2016

Split-brain auto-encoders



Zhang *et al.* CVPR 2017

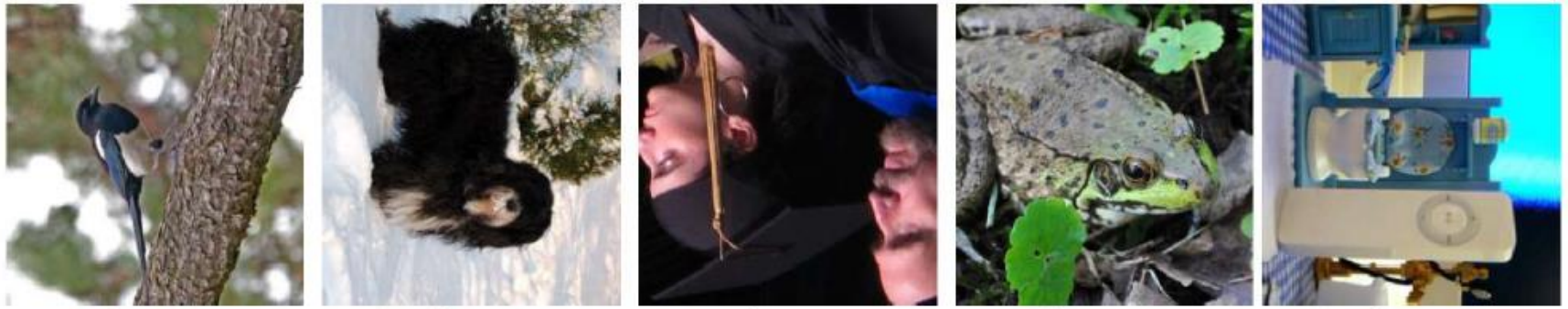
Self-Supervised Learning

State of the art in this line?!

Which image has the correct rotation?



What rotation has been applied to this image?



[Gidaris et al. “Unsupervised representation learning by predicting image rotations”, ICLR 2018]

State of the art in this line?!

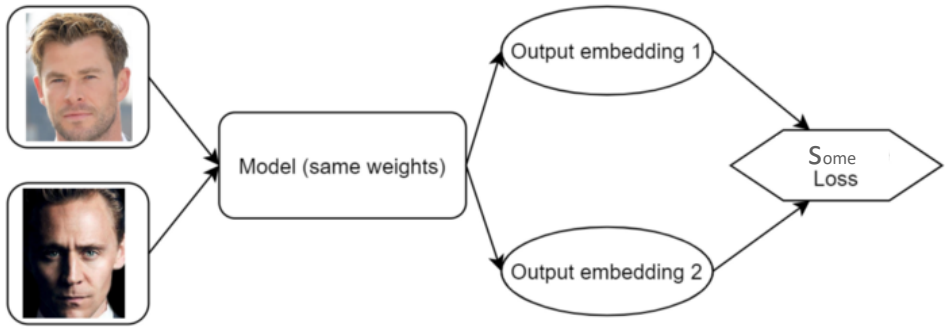
	PASCAL VOC Detection mAP
Random	43.4
Rel. Pos.	51.1
Colour	46.9
Rotation	54.4
ImageNet Labels	56.8

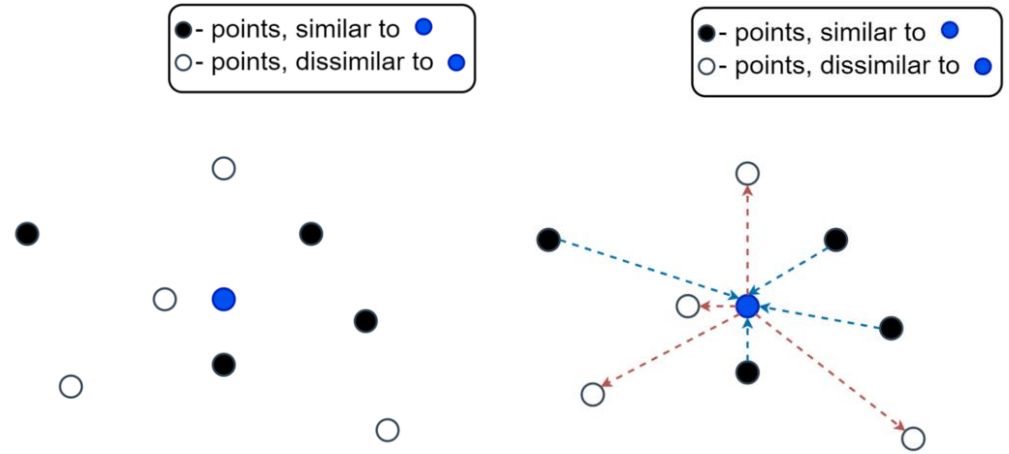
[Gidaris et al. “Unsupervised representation learning by predicting image rotations”, ICLR 2018]

Metric Learning

- Instead of learning to classify a single sample, learn to compare a pair of samples

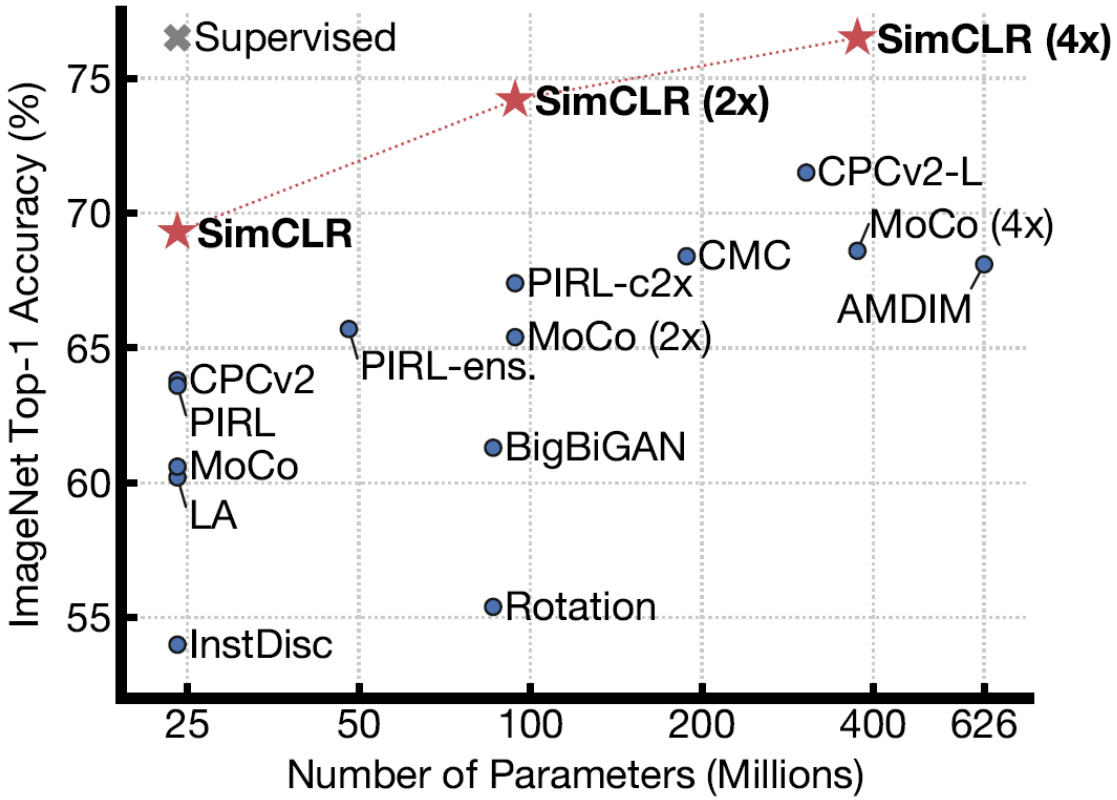
$$\mathcal{L}(\mathcal{D}) = \sum_{(\mathbf{x}_i,\mathbf{x}_j)\in similar} l_s\left(d\left(f_\theta(\mathbf{x}_i), f_\theta(\mathbf{x}_j)\right)\right) + \sum_{(\mathbf{x}_i,\mathbf{x}_j)\in dissimilar} l_{ds}\left(d\left(f_\theta(\mathbf{x}_i), f_\theta(\mathbf{x}_j)\right)\right)$$





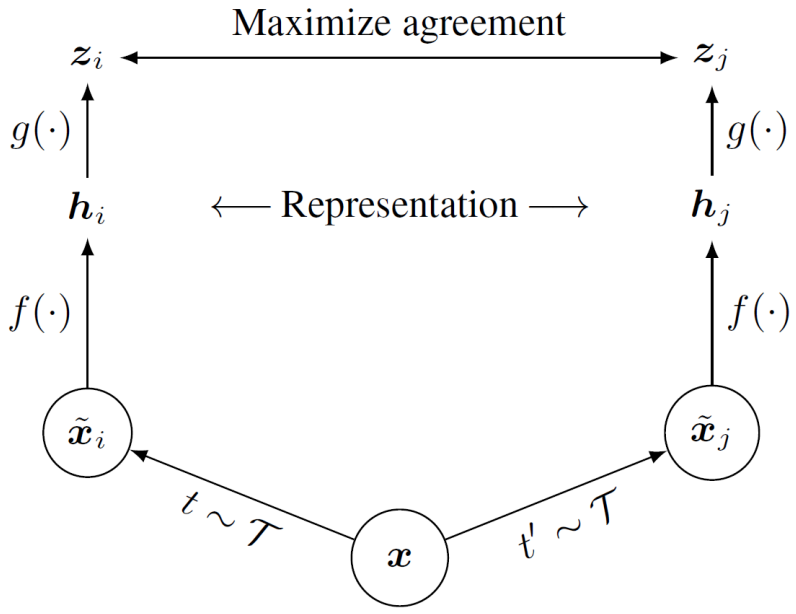
[Images taken from https://medium.com/@maksym.bekuzarov/losses-explained-contrastive-loss-f8f57fe32246]

Example 3: SimCLR (Constrastive Loss)



[Chen et al. "A Simple Framework for Contrastive Learning of Visual Representations", ICML 2020]

Example 3: SimCLR (Constrastive Loss)

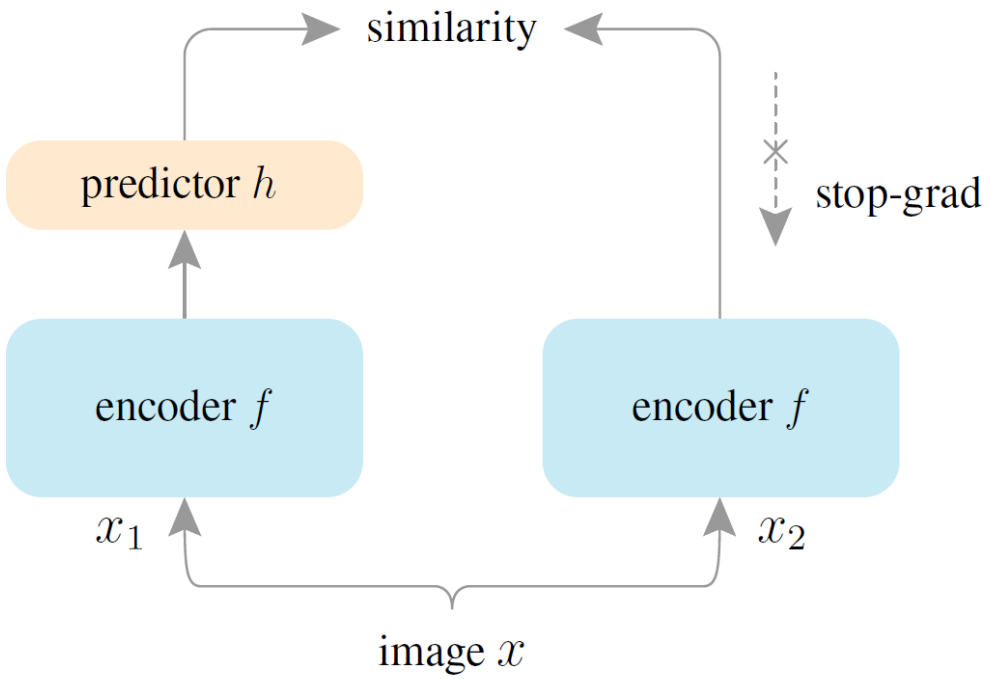


$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)}$$



[Chen et al. "A Simple Framework for Contrastive Learning of Visual Representations", ICML 2020]

Example 3: Learning with only similar pairs



[Chen and He. "Exploring Simple Siamese Representation Learning", CVPR 2021]
[Grill et al. "Bootstrap your own latent: A new approach to self-supervised Learning", NeurIPS 2020]